BA-TOS updates

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Where we started: the SIBGRAPI 2026 result

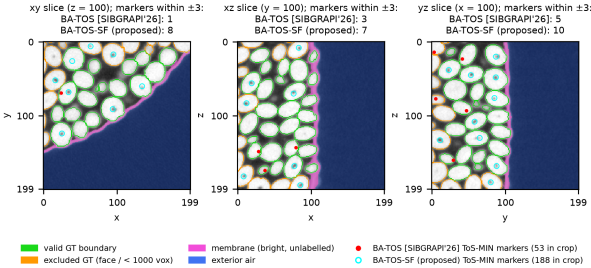
Data	Crops	Specimens	Crop-macro F1@0.50	Without safeguard
EFRGP01, 2 scans, 200^3	12	1	**0.906** $\pm$ **0.210** (min 0.260)	0.824 $\pm$ 0.367 (min 0.012)

Three open issues

1. **Support collapse on two crops** handled by a safeguard; cause not yet identified.
2. **Parameters:** many constants, sensitivity measured on one crop only.
3. **No external validation:** one specimen, the same 12 crops used for design and evaluation.

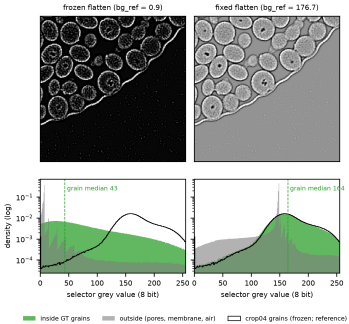
Source: `results/batos_sibgrapi/crop_macro_summary.csv (seed-ignore, no size filter, sample SD)`

Crop05 failure explained: exterior air biases the flatten offset



Membrane (magenta), exterior air (blue, 51 % of the crop); markers:

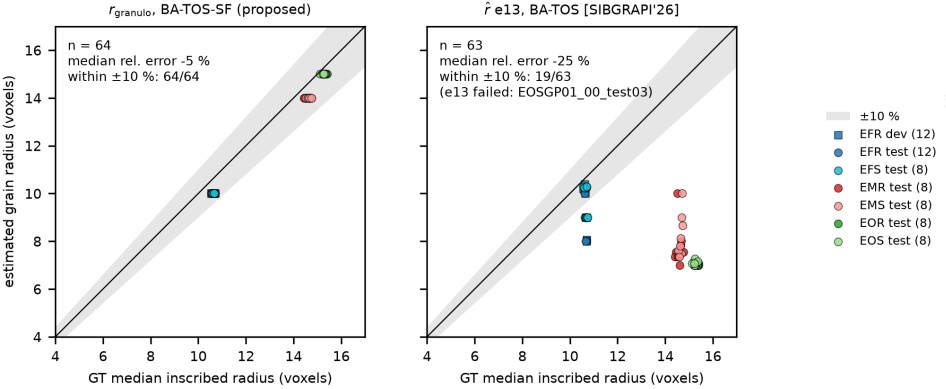
SIBGRAPI (red) vs BA-TOS-SF (cyan)



Offset: 0.9 (whole crop) vs 176.7 (specimen only)

- Air at value 0 drags $\mathrm{median}(\mathit{blur} > 0)$ to 0.9; **one-line change:** $\mathrm{median}(\mathit{blur} \geq 1)$.
- With the offset computed on the specimen only, the collapse does not occur and the safeguard is no longer needed; crop-macro F1@0.50 **0.824** $\rightarrow$ **0.976**.

Scale: a granulometric estimator with no parameter



- r_{granulo} = argmax of the pattern spectrum of the specimen foreground: **64/64 crops within $\pm$10 %** of the GT radius.
- The SIBGRAPI estimator (e13): 19/63 within $\pm$10 %; it **halves the radius** on the larger EM / EO grains.

BA-TOS-SF (v3): what changed

- Scale from r_{granulo}; every size constant follows $s = r_{\mathrm{granulo}}/10$.
- Thresholds from specimen-only statistics (Otsu inside the specimen).
- Absolute grey-level guards removed.
- Safeguard no longer needed (never activates on 52 test crops).

93 constants	SIBGRAPI	BA-TOS-SF
Fixed in voxels (size, volume, count)	43	**0**
Ad-hoc thresholds	6	**0**
Absolute grey-level guards	3	**0**
Dimensionless (kept)	41	41
Safeguard activates (52 test crops)	30	**0**

Size constants scale as s, s^3 or $1/s^3$; on the dev crops $s = 1$.

Source: `docs/V3_SCALE_FREE.md` (full constant table); `paper/tables/test_v3_tables.tex`

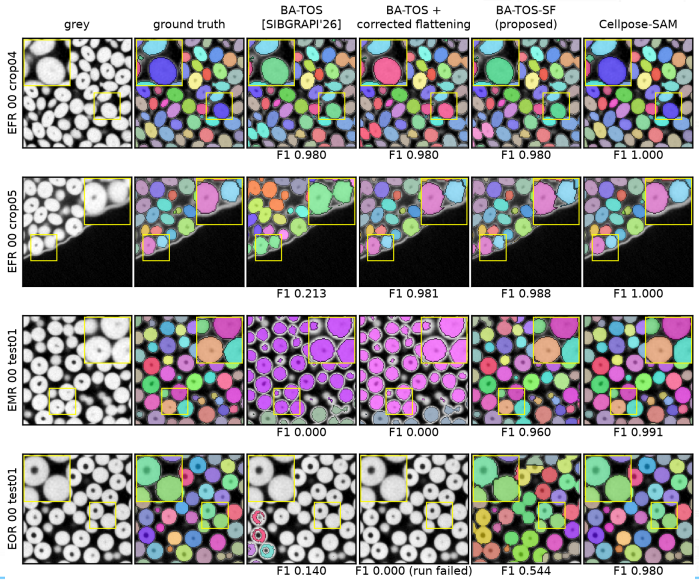
Held-out validation: 52 crops, 6 specimens

Specimen	n	$\hat{r}$ / r_{in}	SIBGRAPI'26	+ specimen-only offset	BA-TOS-SF
EFR	12	10 / 10.6	0.879 ± 0.223	0.956 ± 0.048	**0.985 ± 0.011**
EFS	8	10 / 10.6	0.882 ± 0.155	0.968 ± 0.043	**0.974 ± 0.044**
EMR	8	14 / 14.5	0.303 ± 0.439	0.250 ± 0.416	**0.980 ± 0.027**
EMS	8	14 / 14.7	0.432 ± 0.465	0.479 ± 0.474	**0.986 ± 0.020**
EOR	8	15 / 15.3	0.293 ± 0.362	0.238 ± 0.434	**0.754 ± 0.240**
EOS	8	15 / 15.3	0.265 ± 0.429	0.365 ± 0.499	**0.754 ± 0.214**
External (5 spec.)	40	–	0.435 ± 0.435	0.460 ± 0.473	**0.890 ± 0.178**
All	52	–	0.538 ± 0.437	0.575 ± 0.465	**0.912 ± 0.161**

- Protocol pre-registered: crop list committed before cutting (GT-free rule, fixed seed); nothing tuned on test.
- BA-TOS-SF reaches 0.98 on the EM specimens; 0.75 on the large-grained EO specimens.

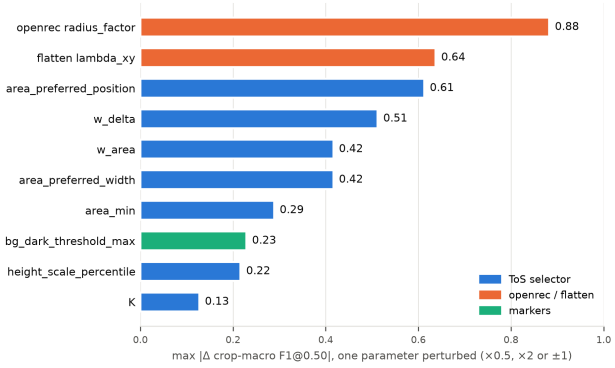
Crop-macro F1@0.50, seed-ignore, no size filter. Source: `results/test_v3/`

Qualitative: 4 crops × 6 columns



Parameter audit: 96 constants, 25 influential

- 96 method constants swept one at a time (12 dev crops); **25** move F1 by > 0.01.
- Selector weights sit on a **narrow ridge**: halving or doubling one drops F1 by 0.4–0.6.
- `w_support` has **no effect**: $\Delta = 0.000$.



Source: `results/sensitivity/ranking.csv` (top 10 method parameters, BA-TOS_sel route)

Comparison with Cellpose-SAM (default settings)

Specimen	`cpsam_v2`	BA-TOS-SF
EFR	**0.999**	0.977
EFS	**0.999**	0.964
EMR	**0.998**	0.966
EMS	**0.999**	0.970
EOR	**0.996**	0.735
EOS	**0.998**	0.727
All 52	**0.998**	0.896

52 test crops	F1@0.50	mIoU
Cellpose-SAM	0.998	0.899
Cellpose, eroded 1 voxel	**0.999**	0.917
Hybrid	0.996	**0.923**

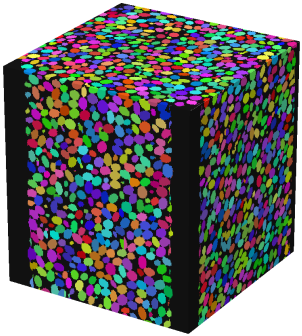
Crop-macro F1@0.50, no-marker evaluation (same rule for all).

Hybrid = Cellpose markers + BA-TOS-SF watershed.

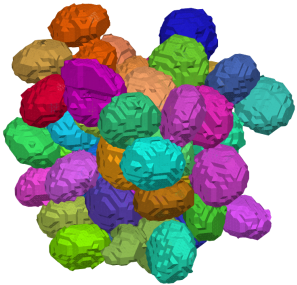
- **Settings:** one default config, no fine-tuning; all four Cellpose 4 models agree within 0.002.
- **Robustness:** at 5 % added noise BA-TOS-SF drops to 0.13 / 0.06 / 0.00 (3 crops), `cpsam_v2` $\geq$ 0.977; at 2$\times$ scale 0.963 vs 0.900.
- **Training data:** documented set = 19 cell/nuclei datasets, 2D (plus 98 natural photos); no micro-CT, granular data or Zenodo 8014905. The undocumented June 2026 models score the same as the documented `cpsam`.
- **Boundaries:** a 1-voxel erosion of Cellpose-SAM gives mIoU 0.917; the hybrid gives 0.923.

Backup: full-volume tiled BA-TOS-SF

(a) stitched BA-TOS-SF (proposed) block (600^3 core, 3×3×3 tiles), faces cut 30 voxels inside (540^3 shown)



(b) 55 grains with centroid in a central 100^3 sub-block (marching cubes)



- Central $3 \times 3 \times 3$ tiles of EFRGP01_00: 600^3 core, halo 20, orphan-rescue stitching.
- 9,753 instances; F1@0.50 = 0.865 on the 6,005 valid GT grains (seed-ignore, no filter).

The end

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QUESTIONS AND COMMENTS?



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